ASO Atlas 2.0: Evaluating antisense oligonucleotide prediction across the preclinical pipeline

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Abstract

Predictive models are increasingly used to prioritise preclinical drug candidates, but standard accuracy metrics treat all pipeline stages equally, leaving the true impact of these models on total screening costs unclear. This issue is particularly acute in the context of antisense oligonucleotides (ASOs), where producing a single development candidate currently requires screening hundreds of sequences. We introduce a cost-based evaluation framework that translates per-stage classifier enrichment into projected cost savings under a representative preclinical pipeline model. Alongside this framework, we present ASO Atlas 2.0, the largest public multi-endpoint ASO preclinical dataset, spanning 274,280 in vitro efficacy measurements and 20,727 in vivo toxicology measurements across 430 target genes. We find that incorporating model predictions across all stages reduces projected pipeline cost by 70.4%. This saving is largely attributable to modest enrichment at costly late-stage in vivo studies rather than strong enrichment at lower-cost in vitro assays, reversing the model ranking suggested by accuracy alone and highlighting the need for improved toxicology modelling.

1 Introduction

Preclinical drug development screens candidate compounds through sequential stages of increasing cost and complexity, from in vitro assays to animal toxicity studies, with late-stage failures dominating total programme expenditure [1, 2]. Predictive models are increasingly used to prioritise candidates in this process, but are typically developed and evaluated on individual assay endpoints in isolation. Because each stage acts as a filter, pass rates compound multiplicatively: a higher failure rate at any stage forces more candidates into every preceding stage, so total programme cost is governed by the product of stage-wise attrition rates, not by any single stage in isolation [3]. This makes it impossible to assess the value of improved prediction at one stage without modelling its interaction with every other stage.

We study this disconnect in the context of antisense oligonucleotides (ASOs), short synthetic nucleic acids that hybridise to target RNA to modulate gene expression [4]. In recent years, ASOs have emerged as a leading therapeutic modality for central nervous system disorders, with the approved

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drug tofersen for *SOD1*-linked ALS [5] and multiple programmes in late-stage trials [6, 7]. Yet producing a single development candidate still requires empirical screening of hundreds of designs [8, 9]. Candidates progress through in vitro efficacy and potency assays, then hepatic and neurological toxicity studies in mouse, rat, and non-human primate, a pipeline in which per-compound costs escalate sharply at each stage. [10]

Existing ASO datasets and models each address individual endpoints in isolation, making it impossible to assess whether improved prediction at one stage reduces total pipeline cost. To address this gap, we release ASO Atlas 2.0, the largest public multi-endpoint ASO preclinical dataset, extracted from Ionis Pharmaceuticals' USPTO patent filings and annotated with Hierarchical Editing Language for Macromolecules (HELM) [11] chemical-structure strings. The dataset spans 274,280 in vitro efficacy and 20,727 in vivo toxicology measurements across 430 target genes and four assay types. We pair it with a cost-based evaluation framework that translates per-stage classifier enrichment into projected pipeline cost savings and animal reduction, and benchmark seven sequence-based architectures.

2 Related work

Computational modelling of ASO properties. Existing methods have addressed individual endpoints in isolation. Early efficacy prediction relied on sequence motifs correlated with antisense activity [12] and hybridisation thermodynamics, as in OligoWalk [13]. More recently, ASOptimizer combines a linear regression on thermodynamic features for sequence selection with a graph neural network for chemical modification optimisation, but treats the two as independent problems [14]. Our previous model, OligoAI, jointly encodes sequence and chemistry to achieve 3.14$\times$ enrichment for in vitro knockdown [15]. For toxicity, nucleotide composition has been shown to predict hepatotoxicity in LNA gapmers [16, 17], and this approach was extended to neurotoxicity [18]. Each of these methods targets a single assay type; none has been evaluated across the full preclinical pipeline.

Benchmarking oligonucleotide and molecular property prediction. Our previous dataset, ASO Atlas, comprises 188,521 in vitro efficacy measurements [15], and OligoGym is the first standardised benchmark suite for oligonucleotide property prediction, spanning multiple endpoints including efficacy, toxicity, and pharmacokinetics [19]. However, OligoGym's datasets are drawn from independent sources with no shared compounds between endpoints, precluding cross-stage analyses of how prediction at one endpoint affects outcomes at another. In the broader drug discovery literature, shared benchmarks such as ChEMBL [20], MoleculeNet [21], and Therapeutics Data Commons [22] have demonstrated how standardised evaluation accelerates method development.

Economics of personalised ASO therapy. In recent years, individualised ASO therapies have demonstrated clinical proof-of-concept for patients with ultra-rare mutations [23], and frameworks for N-of-1 trials are now being established [24, 25, 26]. However, each individualised programme must still screen hundreds of candidates through the full preclinical pipeline [8]. Scaling this approach to the millions of patients with ultra-rare mutations will require drug discovery that is "rapid, automated and cost-effective" [27]. Computational models that reduce the number of candidates entering expensive late stages could therefore have outsized impact on the feasibility of personalised programmes, yet no framework translates preclinical-stage classifier enrichment into projected cost savings.

3 ASO Atlas 2.0

ASO Atlas 2.0 is the first public ASO dataset to link chemical structure to multiple preclinical endpoints across the screening pipeline. Prior public resources cover individual assays in isolation, which precludes the cross-stage analyses needed to evaluate sequential screening. The dataset comprises four linked assay categories extracted from USPTO patent filings: 174,459 single-concentration in vitro inhibition measurements across 163,802 ASOs, 99,821 multi-dose response measurements across 16,834 ASOs, 15,831 hepatorenal toxicity measurements across 2,051 ASOs with 7 biomarkers, and 4,896 neurotoxicity measurements across 3,316 ASOs (Figure 1). All compounds are annotated with HELM chemical-structure strings enabling extraction of chemistry-aware features.

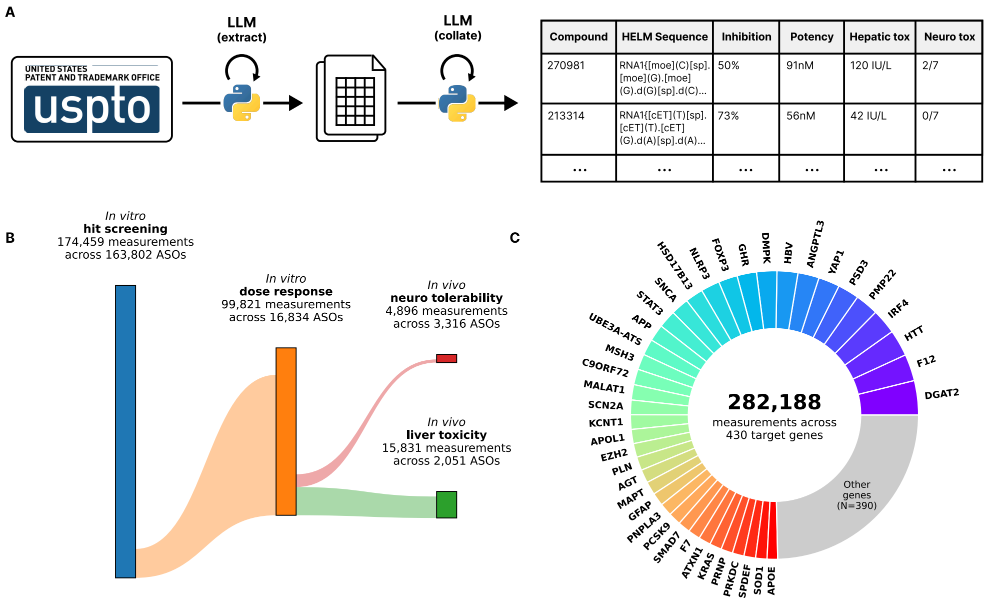

Figure 1: Overview of ASO Atlas 2.0. **(A)** Three-stage LLM-powered pipeline converting USPTO patent tables into structured preclinical datasets with HELM annotations. **(B)** Sankey diagram of compound overlap between assay stages; box heights are proportional to measurement count and flow heights are proportional to the fraction of compounds shared between adjacent stages. **(C)** Donut chart of measurement distribution across 430 target genes. The 40 largest genes are labelled individually and account for 75% of all measurements; the remaining 390 genes are grouped as "Other".

3.1 Extraction Pipeline

ASO Atlas 2.0 was constructed by a three-stage language-model-powered pipeline that converted unstructured tables in USPTO patent XML files into flat, structured preclinical ASO datasets annotated with HELM chemical-structure strings. The full pipeline used GPT-5-mini (Stages 1–2) and GPT-5 (Stage 3) via the OpenAI API at an approximate total cost of $500.

Stage 1 (table discovery). Patent XML documents were retrieved, split into individual tables, deduplicated, and reduced from 35,871 raw tables across 1,125 Ionis Pharmaceuticals patents to 8,435 canonical tables. For each table, GPT-5-mini generated a bespoke Python extraction function executed in a sandboxed subprocess with an agentic self-repair loop (up to five attempts per table).

Stage 2 (chemistry annotation). The model generated functions to construct HELM chemical-structure strings nucleotide-by-nucleotide from the patent prose, returning null when explicit modification data could not be found. All HELM strings were validated by a rule-based checker enforcing correct sugar, base, and backbone tokens, balanced bracket syntax, and terminal-nucleotide constraints.

Stage 3 (assay extraction). Each assay category was extracted by supplying a target schema; GPT-5 generated per-table mapping functions that translated heterogeneous field names, performed unit conversions, and unpivoted multi-measurement rows. HELM annotations were merged by compound identifier, following canonical-link chains across duplicate tables. Full collation details and schema definitions are given in Section A.1 and Section A.5.

Each assay category underwent standardised HELM-level quality filtering (removing uncertain annotations, short sequences, non-gapmer designs, homopolymers, and unmodified DNA) followed by assay-specific range filtering, unit harmonisation, species/cell-line standardisation, and deduplication. Target RNA names were mapped to canonical HGNC gene symbols via the Ensembl REST API supplemented by a manually curated alias dictionary. Full preprocessing details are given in Section A.2. Dataset characterisation, including cross-endpoint correlations and cross-species concordance, is reported in Section A.3.1.

As ASO Atlas 2.0 was constructed by LLM-driven extraction rather than manual curation, we performed a manual validation audit to quantify end-to-end accuracy. A stratified random sample of 100 rows was drawn across all four assay categories, and each was independently verified against its source patent on three axes: (i) HELM chemical-structure string (correct sugar, backbone, and base annotations matching the patent's modification table), (ii) chemistry metadata (species, cell line, dosage, and administration route), and (iii) primary data point (the reported assay value (percent inhibition, IC50, biomarker level, or FOB score)). **[TODO: insert accuracy figures once audit is complete]**

4 Evaluation Framework

4.1 Pipeline Structure and Cost Model

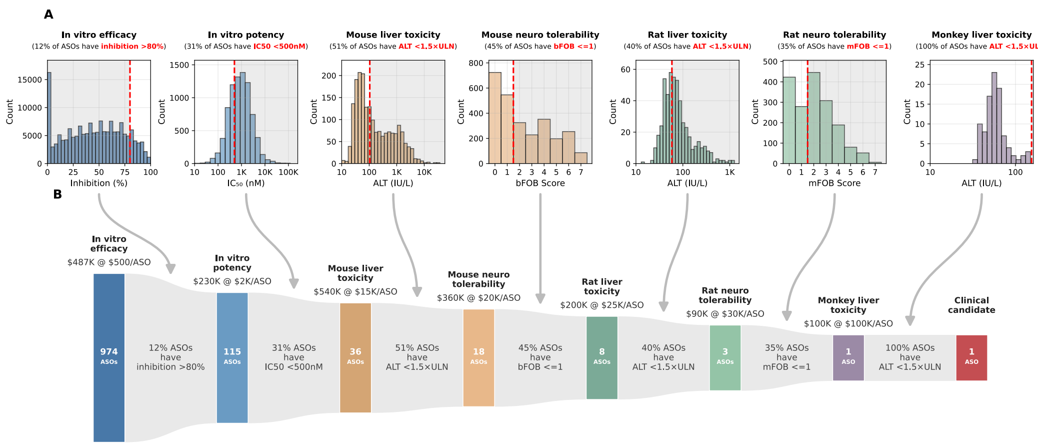

Figure 2: The ASO preclinical screening pipeline across 7 sequential stages. **(A)** Histograms of assay readouts observed in ASO Atlas 2.0 at each stage; red dashed lines indicate the pass/fail threshold used to define stage advancement, grey shading marks the failing region. **(B)** Attrition funnel back-calculated from ASO Atlas 2.0 pass rates: 974 initial ASOs are screened to yield one clinical candidate at an illustrative total cost of $2M under representative pricing assumptions. Box heights are proportional to log(ASO count); annotations show per-stage pass rates and cumulative costs.

We model the ASO preclinical pipeline as seven sequential stages, each with a representative pass/fail threshold: in vitro efficacy (inhibition >80%), in vitro potency (IC50 <500 nM by electroporation), mouse hepatotoxicity (ALT <1.5× the upper limit of normal (ULN)), mouse neurotoxicity (bFOB ≤1), rat hepatotoxicity (ALT <1.5×ULN), rat neurotoxicity (mFOB ≤1), and monkey hepatotoxicity (Figure 2). The sequential ordering follows standard industry practice [8]; pass rates at each stage are computed from ASO Atlas 2.0 by applying these thresholds to the observed assay readouts. Back-calculation from the observed pass rates yields approximately 974 initial ASOs required to produce a single development candidate, at an estimated total cost of $2.01M. Per-ASO costs, estimated from CRO pricing schedules and expert consultation, increase by roughly two orders of magnitude across the pipeline: $500 (in vitro efficacy), $2,000 (dose-response), $15,000 (mouse hepatotoxicity), $20,000 (mouse neurotoxicity), $25,000 (rat hepatotoxicity), $30,000 (rat neurotoxicity), and $100,000 (monkey hepatotoxicity).

To quantify a model's value at a given stage we measure how much it can improve the pass rate when used to prioritise which compounds to advance. We evaluate this at the budget-neutral operating point: compounds are ranked by predicted score and the top $K = q_k N$ are selected, where q_k is the base pass rate and N is the number of compounds evaluated. This selects the same number of compounds as would pass under brute-force screening, so any improvement reflects model quality rather than a change in screening budget. The enrichment factor is then $\mathrm{EF}_k = P(\text{pass} \mid \text{selected})/q_k$: the ratio of the observed pass rate among the K model-selected compounds to the base rate. Random selection yields $\mathrm{EF} = 1$; perfect prediction yields $\mathrm{EF} = 1/q_k$ (every selected compound passes).

Because the effective pass rate under model pre-screening is $\tilde{q}_k = q_k \cdot \mathrm{EF}_k = P(\text{pass} \mid \text{selected})$, the enrichment factor directly multiplies the stage pass rate in the cost model with no additional parameters.

Pipeline costs are back-calculated by determining the number of initial ASOs N_0 needed to yield one development candidate. For a pipeline with S sequential stages, each with pass rate q_k, the number of ASOs entering stage k is $N_k = \lceil N_{k+1}/q_k \rceil$ (working backwards from $N_{S+1} = 1$), and the total cost is $C = \sum_{k=1}^{S} N_k r_k$, where r_k is the per-ASO cost at stage k. With classifier pre-screening, the effective pass rate at enriched stages becomes $\tilde{q}_k = \min(q_k \cdot \mathrm{EF}_k, 1)$.

4.2 Assumptions

The cost model makes several assumptions. For each, we describe the assumption and the evidence or analysis that mitigates it.

(i) Fixed stage ordering. The pipeline assumes a single canonical progression (in vitro efficacy, potency, mouse hepatotoxicity, mouse neurotoxicity, rat hepatotoxicity, rat neurotoxicity, monkey hepatotoxicity). ASO preclinical screening is increasingly performed as a linear stage-gate process: n-Lorem's platform screens ~500 candidates through sequential in vitro potency, selectivity, cyto-toxicity, immunotoxicity, and in vivo tolerability gates before selecting a single clinical candidate [28, 29], and a similar sequential safety-filtering workflow has been described for ASOs more broadly [8]. Our model omits stages present in these workflows for which ASO Atlas 2.0 lacks data, notably immunotoxicity screening; actual workflows may also include parallel assays, iterative design cycles, and organisation-specific orderings.

(ii) Marginal pass rates as proxies for conditional rates. The back-calculation multiplies per-stage pass rates as if each stage screens an independent, correctly filtered population. In practice, pass rates are computed as marginal rates on the observed assay data, implicitly assuming that compounds tested at a later stage (e.g. rat) have already been filtered through all preceding gates. The patent data do not explicitly confirm prior-stage outcomes, but the selection-bias analysis in Figure A1 A provides direct evidence for pipeline survivorship. At each gate, the biomarker distribution of compounds that advance to the next stage is systematically shifted relative to the full population: compounds progressing to dose-response have higher maximum inhibition; those reaching in vivo testing have lower IC_{50} and, most critically for the cost model, compounds tested in rat show lower mouse ALT and lower mouse bFOB scores than those not tested in rat, indicating that mouse-toxic compounds were filtered before rat studies. If compounds were tested in rat irrespective of mouse outcome, these distributions would overlap; the observed shifts confirm that the rat-tested population has already passed through mouse-based selection. Two further observations support this: 90% of rat hepatic and 82% of rat neurotoxicity compounds also appear in the mouse data, confirming high cross-stage overlap; and the monkey stage shows a 100% pass rate (n = 97), consistent with prior-stage filtering removing toxic compounds before primate studies.

(iii) Threshold choices. Each stage uses a single fixed pass/fail threshold. Hepatotoxicity thresholds are defined relative to the upper limit of normal (ULN), taken as the sex-averaged 97.5th percentile of ALT in healthy animals from published reference interval studies (mouse C57BL/6N: 70 IU/L [30]; rat Sprague-Dawley: 39 IU/L [31]; cynomolgus macaque: 103 IU/L [32]). ASOs with ALT <2 $\times$ULN have been classified as having no or very low hepatotoxic potential and those >5$\times$ULN as clearly hepatotoxic [16]; 1.5-fold has been identified as the lower boundary at which any hepatotox-icity signal becomes detectable [33]; our threshold of 1.5$\times$ULN therefore selects only compounds with no detectable transaminase elevation. Neurotoxicity is scored using Ionis's 7-criterion binary checklist [34], where each criterion assesses a functional domain (consciousness, motor response, reflexes) and the total score ranges from 0 (no signs) to 7 (all criteria failed); a threshold of ≤1 permits at most one failed criterion. These thresholds are deliberately conservative; for instance, ION582, a clinical-stage ASO currently in Phase 3 trials for Angelman syndrome, has a reported mFOB score of 2 and would be eliminated under our FOB ≤ 1 threshold despite having progressed through preclinical development. ION582 nonetheless ranks favourably on both potency and toler-ability within the dataset's *UBE3A-ATS*-targeting ASOs (Figure A2), consistent with its selection as a clinical candidate and providing a sanity check that ASO Atlas 2.0 captures clinically relevant signal.

(iv) Fixed dosing conditions. Neurotoxicity pass rates are computed at specific dose–latency combinations (mouse: 700 μg ICV, 3 h; rat: 3000 μg, 3 h), the standardised conditions under which FOB is assessed; compounds tested at other doses are excluded, reducing sample size at these stages.

(v) Cost estimates. Per-ASO stage costs are representative estimates from CRO pricing quotes and expert consultation, not observed programme expenditure. The sensitivity analysis (Figure A4) sweeps all costs $\pm$50% and confirms the headline cost-reduction conclusion is robust to moderate perturbation, but absolute dollar estimates should be interpreted as illustrative.

(vi) Cross-stage enrichment independence. The cost model applies per-stage enrichment factors independently, implying that enriching the population at an upstream stage does not alter the EF achievable at downstream stages. This requires that filtering at one stage does not systematically shift the outcome distribution at subsequent stages. The cross-assay correlation matrix (Figure A1 B) provides direct evidence for independence between in vitro and in vivo stages: Spearman correlations between efficacy metrics and toxicity endpoints are non-significant (max inhibition vs ALT: ρ = 0.0225, n = 1269 max inhibition vs bFOB: ρ = 0.014, n = 937 IC_{50} vs bFOB: ρ = −0.0985, n = 236), with the exception of a weak IC_{50} vs ALT association (ρ = −0.183, n = 954), so enriching for in vitro efficacy leaves the toxicity distribution of the surviving pool largely unchanged. Independence between hepatic and neurotoxicity stages cannot be directly tested: only 21 compounds have both hepatic and neurotoxicity measurements, too few to estimate the correlation reliably, though independence is expected given the distinct biological mechanisms. Within the toxicity stages, cross-species neurotoxicity concordance is strong (ρ = 0.647, Figure A1 B, E), meaning that enrichment at the mouse neurotoxicity stage would reduce the marginal value of the rat neurotoxicity stage; the cost model does not capture this interaction and therefore likely underestimates the combined savings from sequential same-endpoint stages. Cross-species hepatotoxicity concordance is weaker (ρ = 0.299), so this effect is smaller for the hepatic stages.

(vii) Dosage as a confounding factor. Because dosage varies across experiments and is included as a model input for both OligoAI and OligoGym, a model could achieve high EF by ranking lower-dose compounds first rather than learning genuine sequence-level signal. To control for this, enrichment factors at stages with variable dosage are computed with within-dosage stratification: compounds are ranked within each dosage group, the top-K_g are selected per group ($K_g = \text{base rate}_g \times n_g$, budget-matched within each stratum), and all selections are pooled before computing the overall EF, ensuring any enrichment reflects sequence-level prediction quality.

5 Benchmark

All models were evaluated using GroupKFold cross-validation with 5 folds, where groups were defined by USPTO patent ID with HELM-level deduplication (each unique HELM sequence is assigned to its earliest patent, so no ASO appears in both train and test).

5.1 Models

Hagedorn-Linear. We replicated the [18] linear model for predicting ASO neurotoxicity from sequence features, evaluated on mouse bFOB scores (700 μg intracerebroventricular (ICV); neurotoxic bFOB > 1 vs non-toxic bFOB ≤ 1) across 2,696 compounds and 23 target gene groups. The original 5-feature model uses G, A, T, and C nucleotide counts plus the G-free stretch from the 3′ end; fixed coefficients from the original R code were applied directly (no retraining). Compounds with mean bFOB > 1 were labelled “neurotoxic”; those with mean bFOB ≤ 1 were labelled “non-toxic”. As with all other models, enrichment factors were computed by ranking compounds by the continuous predicted score and selecting the top K, without thresholding.

OligoGym benchmark. We benchmarked six model architectures from OligoGym on the six ASO Atlas 2.0 endpoints. Each model was trained in regression mode using OligoGym’s default one-hot featurizer encoding base, sugar, and phosphate components [19] and a single hyperparameter configuration per model. 5-fold GroupKFold cross-validation is grouped by patent with HELM-level deduplication so no ASO sequence appears in both train and test across any fold. The full benchmark ran on a single CPU in approximately 4 hours.

OligoAI. We retrained our previous efficacy model [15] on the ASO Atlas 2.0 in vitro tables under the same 5-fold patent GroupKFold with HELM-level dedup used for the OligoGym benchmark (training details in Section A.4.1).

5.2 Results

Neurotoxicity prediction proved substantially more tractable than hepatotoxicity across all model classes. For mouse bFOB, the best single-task models achieved Spearman ρ = 0.734 ± 0.025, while hepatic ALT prediction remained challenging (ρ = 0.444 ± 0.117). Gradient-boosted methods (CatBoost, XGBoost) and tree ensembles yielded the highest per-stage success rates in the pipeline cost model (Table 1); hepatotoxicity prediction remains difficult for all model classes, consistent with species-specific sequence-toxicity mappings (see Figure A3).

On held-out folds, OligoAI achieved Spearman ρ = 0.474 ± 0.070 for efficacy and ρ = 0.531 ± 0.056 for potency, yielding EF = 4.06 ± 1.17× and EF = 1.55 ± 0.27× respectively.

Table 1: Per-stage success rates and projected pipeline savings for each model. Each cell shows precision@K: the fraction of model-selected compounds that pass the stage, evaluated at the budget-neutral operating point (K = base rate × N). Savings is the projected cost reduction versus the no-screening baseline (974 ASOs, $2.01M). Bold marks the best value per column; the combined row is bolded throughout. Dashes indicate endpoints not covered by that model. Spearman ρ for all models is reported in Table A1.

Model	**Precision@K**						**Savings**
	In vitro		**In vivo**				
			Mouse		Rat		
	Efficacy	Potency	Hepatic	Neuro	Hepatic	Neuro	
Baseline	10.8%	31.0%	57.5%	44.4%	40.5%	35.0%	0.0%
OligoAI	**43.9±12.6%**	**47.9±8.2%**	—	—	—	—	24.3%
Hagedorn-Linear	—	—	—	66.9±9.5%	—	61.2±10.1%	44.1%
Linear	17.7±6.4%	36.9±8.5%	70.6±6.9%	77.2±5.5%	51.4±17.8%	66.3±11.4%	67.4%
Random Forest	20.9±7.1%	39.2±9.7%	70.2±6.9%	77.3±5.2%	**53.2±21.3%**	**68.5±12.4%**	**68.2%**
XGBoost	22.3±3.9%	38.9±9.7%	72.2±6.7%	76.6±6.8%	49.3±20.3%	65.2±14.4%	64.9%
KNN	16.5±5.4%	39.1±6.6%	70.0±7.7%	58.9±10.0%	47.5±24.0%	40.0±14.3%	42.8%
CatBoost	21.9±6.4%	37.5±8.5%	**72.5±6.0%**	**78.6±5.3%**	51.4±22.0%	66.0±12.2%	68.1%
MLP	16.5±5.2%	34.8±9.3%	68.2±7.5%	75.9±6.9%	49.3±20.9%	66.0±13.2%	61.8%
OligoAI + CatBoost (in vivo)	**43.9±12.6%**	**47.9±8.2%**	**72.5±6.0%**	**78.6±5.3%**	**51.4±22.0%**	**66.0±12.2%**	**70.4%**

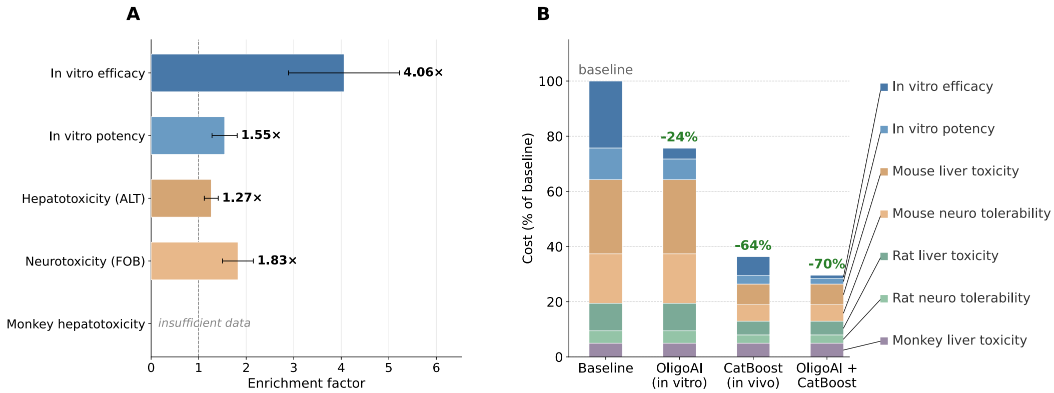

Figure 3: Per-stage enrichment and cost contribution visualising the top strategies from Table 1. **(A)** Per-stage EFs for OligoAI (inhibition), CatBoost (toxicity stages), and their combination. **(B)** Stacked-bar cost decomposition by pipeline stage.

Table 1 compares models on total pipeline cost. OligoAI achieved EF = 4.06 ± 1.17× at efficacy and EF = 1.55 ± 0.27× at potency under 5-fold patent GroupKFold (Section A.4.1); the efficacy EF was higher than the 3.14× we previously reported [15], likely because the model is now trained on both inhibition and dose-response data. The OligoGym architectures trained on ASO Atlas 2.0 cover all six endpoints and compound their enrichment across the pipeline. Per-endpoint CatBoost regressors (one per in vivo stage) give the largest cost savings among the in vivo models in this benchmark, reducing projected cost to \$0.73M, a 63.7% reduction against baseline. Differences between OligoGym architectures are largely within fold-level standard deviations: the practical implication is that any of the gradient-boosted or neural baselines achieves comparable savings, and the choice of CatBoost is robust to within-architecture noise but not strongly preferred over the closest alternatives.

Combining OligoAI (in vitro) with CatBoost (in vivo) yields a total projected cost reduction of 70.4%, though the marginal gain over CatBoost alone is small: because in vitro stages account for only ~35.7% of baseline cost at \$500–2,000 per ASO, even OligoAI's strong enrichment (EF = 4.06 ± 1.17×) translates to modest dollar savings once the expensive in vivo stages dominate (Figure 3 B). This highlights a key insight from the cost-based framework: where a model enriches matters more than how well it enriches; modest improvements at expensive stages outweigh large improvements at cheap ones. Beyond cost, computational pre-screening also reduces animal usage: under the representative pipeline model, CatBoost cuts projected animals from 264 to 92 per development candidate, a 65.2% reduction (Figure A5).

6 Discussion

Existing ASO modelling efforts have focused predominantly on in vitro efficacy prediction, where the largest accuracy gains are achievable. Our cost-based evaluation reveals that this focus is misallocated. While OligoAI achieves substantially stronger per-stage enrichment than CatBoost (EF = 4.06 ± 1.17× vs 1.26–1.88×), CatBoost delivers larger overall cost savings because in vivo stages dominate total programme expenditure. Standard accuracy benchmarks, which treat all endpoints equally, would rank these models in the opposite order, suggesting that current evaluation practice may systematically direct modelling effort away from the stages where it would have the greatest economic impact.

The asymmetry in model performance between neurotoxicity and hepatotoxicity has a mechanistic basis. [35] recently showed that acute neuronal inhibition following CNS delivery is caused by transient high extracellular PS-ASO concentrations that block synaptic transmission via non-specific protein binding; the response peaks 3 h post-dose, reverses within 24 h, and is dose-responsive. Critically, acute inhibition was abrogated in ASOs with lower phosphorothioate and guanine content, establishing a direct compositional determinant of neurotoxicity. This is consistent with the G-content correlation we observe (Figure A1 D) and with the earlier finding that a simple linear model over sequence features predicts neurotoxicity [18]. Hepatotoxicity is also driven by non-specific protein binding of PS-ASOs [36], but the sequence-toxicity relationship is species-specific (Figure A3) and not reducible to base composition alone (Figure A1 D shows no significant base-level signal for hepatic biomarkers). This mechanistic contrast explains why neurotoxicity prediction is substantially more tractable than hepatotoxicity across all model classes.

Since the cost framework identifies hepatotoxicity prediction as the key bottleneck for further pipeline savings, improving it is the most direct path to impact. The models benchmarked here use only chemistry-derived sequence features and do not incorporate off-target hybridisation potential, which has been identified as a major contributor to ASO hepatotoxicity via RNase H-mediated cleavage of unintended transcripts [33, 37]. Multi-task architectures that jointly model in vitro and in vivo endpoints could also exploit the weak but nonzero cross-stage correlations observed in Figure A1 B. Beyond modelling, expanding the hepatotoxicity training data to include additional species and dose levels would help disentangle sequence-intrinsic toxicity from protocol-dependent variation.

These improvements have direct implications for individualised ASO therapy. Organisations such as n-Lorem must screen hundreds of candidates through the full preclinical pipeline for each patient [8], and scaling this to the millions of patients with ultra-rare mutations requires substantially lower per-programme costs [27]. The 70.4% cost reduction and 65.2% animal reduction demonstrated here represent a first step, but the cost framework also clarifies where the ceiling is: the majority of remaining cost is concentrated in the hepatic and neurological safety stages, so closing the hepatotoxicity prediction gap identified above is particularly important for personalised programmes, where preclinical screening cost has been identified as a key bottleneck to broader deployment [27].

The cross-species concordance analysis (Figure A1) provides evidence that ASO Atlas 2.0 reflects real pipeline survivorship: 90% of rat hepatotoxicity and 82% of rat neurotoxicity compounds also appear in the mouse data, and at each gate the biomarker distributions of compounds that advance to the next stage are shifted relative to those that do not (Figure A1 A), consistent with sequential filtering. This supports using observed marginal pass rates directly in the cost model.

Several limitations beyond those of the pipeline cost model (detailed in the Assumptions section above) should be noted. ASO Atlas 2.0 is derived entirely from Ionis Pharmaceuticals patents and therefore reflects the design space, chemical modifications, and screening priorities of a single organisation; generalisability to other chemistries (e.g. LNA gapmers) or other companies' screening cascades is untested and would require independent validation. The cost-based evaluation framework itself is general to any sequential screening pipeline, but the specific cost estimates and stage ordering used here are representative rather than observed, and alternative pipeline configurations could alter the relative value of per-stage enrichment. Finally, the binary classification approach (high vs low toxicity) discards compounds with intermediate biomarker values and may oversimplify the continuous dose-response relationship between sequence features and toxicity.

Taken together, ASO Atlas 2.0 and the accompanying cost-based framework provide a foundation for evaluating computational screening methods in terms of their practical impact on drug development. As predictive models improve, particularly for hepatotoxicity, the framework offers a direct way to quantify their value and guide resource allocation across the preclinical pipeline. This is especially relevant for individualised ASO therapy, where reducing the cost and time of per-patient preclinical screening is a prerequisite for scaling to the broader population of patients with ultra-rare mutations.

7 Acknowledgments and Disclosure of Funding

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8 Code and Data Availability

All code, processed datasets, and trained model weights required to reproduce the results in this paper are available at github.com/ASO-Atlas/aso-atlas-2.0. The dataset is hosted on Hugging Face at huggingface.co/datasets/barneyhill/aso-atlas-2 with Croissant 1.1 machine-readable metadata. The best-performing toxicity model (CatBoost, retrained on ASO Atlas 2.0) is deployed alongside OligoAI at sitlabs.org/oligoai.

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A.1 Collation Framework

USPTO patent XML documents filed after 2001 by Ionis Pharmaceuticals were retrieved via the USPTO bulk-data API and split into individual table files, each paired with the five paragraphs of prose immediately preceding the table. Tables were deduplicated in two phases: exact MD5 hash matching, followed by pairwise sequence similarity (Python `SequenceMatcher`, 90% threshold) with length-based pruning; connected components were resolved by depth-first search to select a single canonical file per group, reducing 35,871 raw tables from 1,125 Ionis Pharmaceuticals patents to 8,435 canonical tables.

For each canonical table, GPT-5-mini generated a bespoke Python extraction function tailored to that table's XML layout. Generated scripts were executed in a sandboxed subprocess with a restricted import whitelist (`json`, `re`, `xml.etree.ElementTree`, `math`) and a ten-second timeout. An agentic self-repair loop allowed the model to observe execution output or errors and regenerate corrected code, for up to five attempts per table.

For HELM annotation, the model received the five paragraphs immediately preceding the table in the patent XML, the table XML itself, and five sampled compound entries from Stage 1, then generated a function that constructs HELM strings nucleotide-by-nucleotide from the chemistry description. To avoid hallucinated annotations, the function was required to return null when explicit modification data could not be found in the patent text. All HELM strings were validated by a rule-based checker that enforced correct sugar tokens (`[moe]`, `d`, `r`, `m`, `[cet]`, `[fR]`, `[lna]`), canonical bases (`A`, `C`, `G`, `T`, `U`, `[5meC]`), backbone tokens (`[sp]`, `[am]`, `.`), balanced bracket syntax, and terminal-nucleotide constraints.

For schema-based collation, each assay category was extracted in a separate pass by supplying a schema dictionary that mapped desired column names to natural-language descriptions. GPT-5 generated a mapping function per table that translated table-specific field names to the target schema, performed unit conversions, and unpivoted multi-measurement rows into separate rows. The same sandboxed execution and self-repair loop were applied. HELM annotations were merged into each output row by compound identifier, following canonical-link chains across duplicate tables. Four schemas were defined, one per assay category: (i) in vitro inhibition: percent inhibition, cell line, dosage in nM, transfection method, treatment period, and target gene; (ii) dose response: the same fields with dosage in the original unit (nM or μM); (iii) hepatorenal toxicity: seven serum biomarkers (ALB, ALT, AST, BUN, CREA, TBIL, protein/creatinine ratio), dosage, species, strain, number of doses, dosing period, measurement source, and administration route; and (iv) neurotoxicity: FOB score (bFOB or mFOB), dosage, species, strain, latency, administration method, and score type (Section A.5).

A.2 Preprocessing and Quality Control

Raw tables were parsed into four assay categories: in vitro hit screening (single-concentration percent inhibition), in vitro dose response (multi-dose percent inhibition), in vivo hepatorenal toxicity (serum biomarkers), and in vivo neurotoxicity (functional observational battery scores). Each category underwent a standardised cleaning pipeline described below.

Five HELM-level quality filters were applied uniformly across all four assay categories before any assay-specific cleaning. Compounds whose HELM annotation contained uncertain sugar or backbone tokens (indicated by `?` placeholders in the HELM string) were removed, as were sequences with ten or fewer nucleotides (likely extraction artefacts), uniformly modified compounds with no DNA gap (steric-blocking ASOs outside the scope of gapmer-focused analysis), homopolymer sequences comprising a single repeated nucleotide (extraction artefacts from poly-T or poly-A placeholder annotations), and naked DNA oligonucleotides lacking any sugar modifications (unmodified sequences where the extraction model could not identify the modification pattern).

In vitro inhibition. Inhibition values outside the range [–1000%, 100%] were discarded as extraction artefacts. Cell line names and species labels were standardised to canonical forms (e.g. "A-431" → "A431", "cynomolgus" → "monkey"). Duplicate measurements (defined as identical compound ID and inhibition value) were collapsed, retaining the most recent patent. This yielded 174,459 measurements across 163,802 ASOs.

Dose response. Dosages reported in μM were converted to nM. The same inhibition range filter and species standardisation were applied. Rows with non-positive dosages were removed. Cell line names were harmonised using a lookup table of 121 known aliases. Deduplication on compound, dosage, and inhibition yielded 99,821 measurements across 16,834 ASOs.

Hepatorenal toxicity. Dosage strings were parsed into numeric values and units. Weekly dosing rates (mg/kg/wk) were converted to per-dose equivalents (mg/kg) using the reported number of doses and dosing period. Only subcutaneous and intraperitoneal administration routes with plasma or urine measurement sources were retained. Extreme dosages ($>$ 10,000 mg/kg) were excluded. Biomarker values were range-filtered (ALT and AST: 10–50,000 IU/L; ALB: 1–100 g/dL). Rows sharing the same compound, species, dosing regimen, and administration method were collapsed, aggregating biomarker replicates into lists. Species and strain names were standardised. The final dataset contained 4,341 observations across 2,051 ASOs, with 7 biomarker channels (ALB, ALT, AST, BUN, CREA, TBIL, protein/creatinine ratio).

Neurological toxicity. Tolerability score strings (bFOB for mice, mFOB for rats) were parsed into numeric lists, retaining only values in the valid range [0, 7].

ASO Atlas 2.0 incorporated two complementary tolerability scoring systems developed at Ionis Pharmaceuticals [34]. In each system, seven criteria are each scored 0 (met) or 1 (not met), and summed into a tolerability score ranging from 0 (no signs) to 7 (all criteria failed). The *behavioural FOB* (bFOB) assesses consciousness and motor responsiveness 3 h after intracerebroventricular (ICV) delivery of 700 μg in C57BL/6 mice: bright/alert/responsive, standing or hunched without stimuli, any movement without stimuli, forward movement after lifting, any movement after lifting, response to tail pinch, and regular breathing. The *motor FOB* (mFOB) assesses segmental motor function 3 h after intrathecal (IT) delivery of 3,000 μg in Sprague-Dawley rats: movement of the tail, posterior posture, hind limbs, hind paws, forepaws, anterior posture, and head. In the processed dataset, bFOB accounts for 3,025 mouse observations and mFOB for 1,875 rat observations. Species and strain names were standardised (e.g. "C57/B16 mice" $\rightarrow$ "C57BL/6 mice"). Since the raw neurotoxicity data lacked target gene annotations, compound-to-gene mappings were inferred in two steps: first by direct lookup against the in vitro and dose response datasets, then by patent-level majority vote for remaining compounds (assigning the most frequent target gene within each USPTO patent). This resolved target gene identity for 97% of neurotoxicity observations. Deduplication on HELM annotation, species, administration method, score type, and dosage yielded 4,900 observations across 3,316 ASOs.

Gene symbol mapping. Target RNA names from the patent text were mapped to canonical HGNC gene symbols using the Ensembl REST API, supplemented by a manually curated dictionary of 300+ aliases (e.g. "Tau" $\rightarrow$ *MAPT*, "PKK" $\rightarrow$ *PRKDC*, "K-Ras" $\rightarrow$ *KRAS*). After merging synonyms, the combined dataset spans 430 unique target genes and 282,188 total measurements.

A.3 Dataset Characterisation

A.3.1 Cross-Species Concordance and Biomarker Structure

To assess whether mouse-based screening translates to rat outcomes, we compared per-compound mean biomarker values for compounds tested in both species (Figure A1). For hepatotoxicity biomarkers, cross-species Spearman correlations were modest but statistically significant: ALT (ρ = 0.299, p < 10^{-10}, n = 449), AST (ρ = 0.272, p < 10^{-8}, n = 435), and TBIL (ρ = 0.244, p < 10^{-5}, n = 374).

Neurotoxicity showed stronger cross-species concordance. Mouse bFOB and rat mFOB scores (700 μg ICV vs 3,000 μg intrathecal (IT) respectively) were well correlated (ρ = 0.647, p $\approx$ 0, n = 1,456), with a binary concordance rate of 0.676 across 1456 classifiable compounds.

Within mouse, the hepatotoxicity biomarkers showed a correlation structure consistent with known clinical biochemistry [38] (Figure A1 C). ALT and AST were strongly correlated (ρ = 0.92), as expected given that both are cytoplasmic aminotransferases released upon hepatocyte damage. TBIL was moderately correlated with both ($\rho \approx 0.4$), reflecting its partially distinct mechanistic origin. The renal markers BUN and CREA formed a separate cluster (ρ = 0.36), largely independent of the liver enzymes. This structure confirms that ALT and AST carry largely redundant information for toxicity classification, while TBIL and renal biomarkers may offer complementary signal.

Base composition analysis (Figure A1 D) revealed that guanine content was strongly positively correlated with neurotoxicity scores in both species (mouse bFOB $\rho = 0.602$, rat mFOB $\rho = 0.543$). For hepatic biomarkers, cytosine content was positively correlated with rat ALT ($\rho = 0.236$) but not mouse ALT ($\rho = -0.0335$, n.s.), consistent with the modest cross-species ALT concordance ($\rho = 0.299$) reported in Figure A1 B. Figure A1 E visualises the cross-species neurotoxicity concordance as an integer-rounded score heatmap, showing strong diagonal concentration consistent with the quantitative correlation reported above.

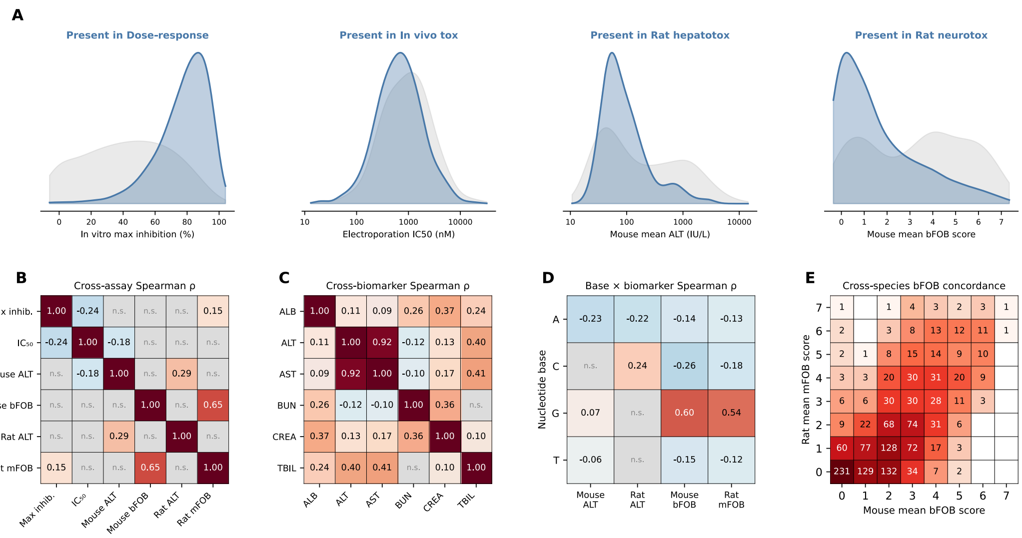

Figure A1: Selection bias, cross-assay and cross-species correlations, biomarker structure, and base composition. **(A)** Selection-bias KDEs showing biomarker distributions for compounds that do (blue) vs don't (grey) advance to the next pipeline stage. **(B)** Cross-assay and cross-species Spearman ρ: pairwise correlations between per-compound metrics across pipeline stages and species (BH-corrected; n.s. = not significant). **(C)** Cross-biomarker Spearman ρ: correlations between mouse hepatotoxicity biomarkers (Bonferroni-corrected). **(D)** Nucleotide base $\times$ biomarker Spearman ρ: base composition vs toxicity biomarkers (BH-corrected). **(E)** Cross-species bFOB concordance: mouse bFOB vs rat mFOB score heatmap (integer-rounded); cell values are compound counts.

A.3.2 Clinical Validation

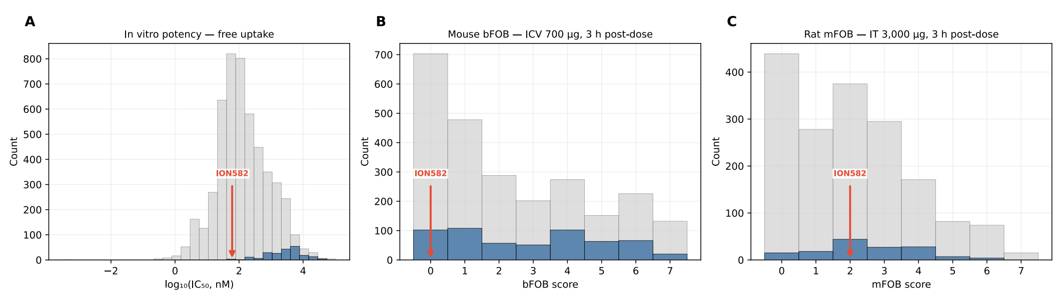

Figure A2: Clinical validation: ION582 (ACCATTTTGACCTTCTTAGC, 5-10-5 MOE gapmer), a Phase 3 candidate for Angelman syndrome, in context of ASO Atlas 2.0 distributions. Grey bars show all targets; blue bars highlight *UBE3A-ATS*-targeting ASOs; the red arrow marks ION582. **(A)** IC50 distribution under gymnosis/free-uptake conditions. **(B)** Mouse bFOB score distribution (ICV, 700 μg, 3 h post-dose). **(C)** Rat mFOB score distribution (IT, 3,000 μg, 3 h post-dose).

A.3.3 Cross-Species Transfer and Baseline Models

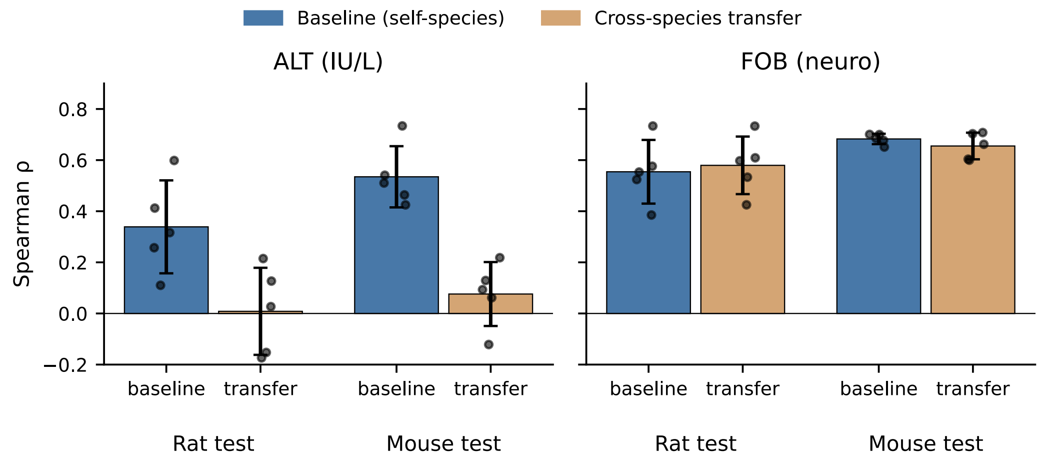

Figure A3: Cross-species transfer of sequence-based toxicity prediction (Random Forest, 5-fold GroupKFold by patent on the test species). Each panel compares baseline (train and test on the same species) against cross-species transfer (train on one species, test on the other). Spearman ρ values are shown per panel.

Table A1: Spearman ρ (mean $\pm$ s.d. across 5 folds) for each model and endpoint. GroupKFold cross-validation grouped by patent with HELM-level deduplication. Bold indicates best model per endpoint.

Model	**Spearman ρ**					
	In vitro		**In vivo**			
	Efficacy	Potency	Mouse		Rat	
			Hepatic	Neuro	Hepatic	Neuro
OligoAI	**.474±.070**	**.531±.056**	—	—	—	—
Hagedorn-Linear	—	—	—	.523±.068	—	.479±.086
Linear	.205±.039	.195±.105	.312±.110	.690±.047	.180±.105	**.619±.092**
Random Forest	.271±.043	.253±.085	.434±.119	.702±.021	.209±.159	.599±.099
XGBoost	.279±.028	.184±.058	.415±.110	.680±.023	.179±.107	.557±.101
KNN	.169±.028	.224±.086	**.444±.117**	.355±.018	**.243±.105**	.245±.116
CatBoost	.319±.045	.192±.072	.414±.148	**.734±.025**	.167±.163	.604±.089
MLP	.180±.035	.177±.155	.442±.172	.700±.021	.223±.144	.616±.108

A.4 Model and Evaluation Details

A.4.1 OligoAI Training Details

Our previous efficacy model, OligoAI [15], a RiNALMo-based sequence+chemistry model for ASO inhibition, was trained on the combined ASO Atlas 2.0 in vitro inhibition and dose-response tables (116,311 efficacy rows from 1798 patent tables). Chemistry was restricted to DNA/MOE/cET backbones with PO/PS linkages (covering 99.8% of parsable rows); human cell lines were retained and target transcripts resolved to GRCh38 via Ensembl release 110 with a $\pm$50 nt flanking context. Rows were deduplicated at the (HELM, cell line, target RNA, dosage) level with mean inhibition. Potency was evaluated as rank correlation of log IC_{50} from 4PL fits to dose-response curves (7979 of 13556 curves converged). Each fold was trained on a single NVIDIA RTX A6000 GPU (RunPod) for approximately 5 GPU-hours (~25 GPU-hours total across 5 folds). Cross-validation procedure and enrichment results are reported in the main text.

A.4.2 Cost Sensitivity and Impact

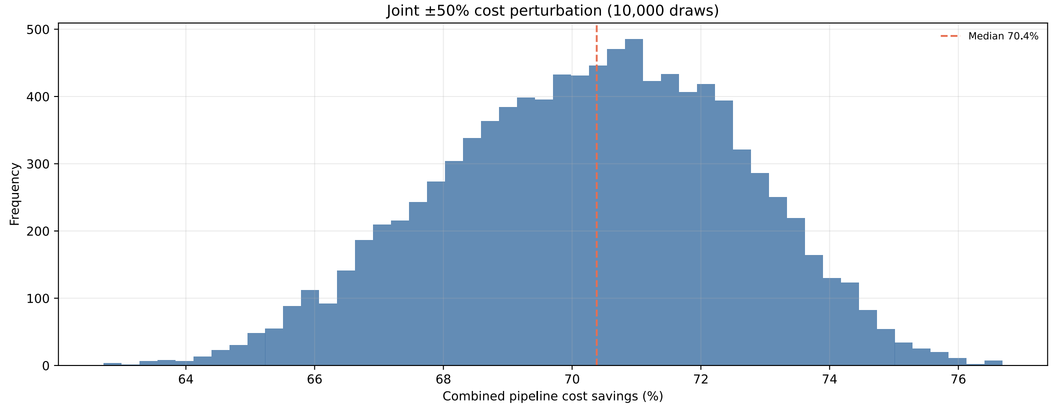

Figure A4: Sensitivity of combined pipeline savings to per-stage cost assumptions. All seven stage costs are drawn independently from $U[0.5\times, 1.5\times]$ their nominal values (10,000 Monte Carlo samples). Dashed line marks the median.

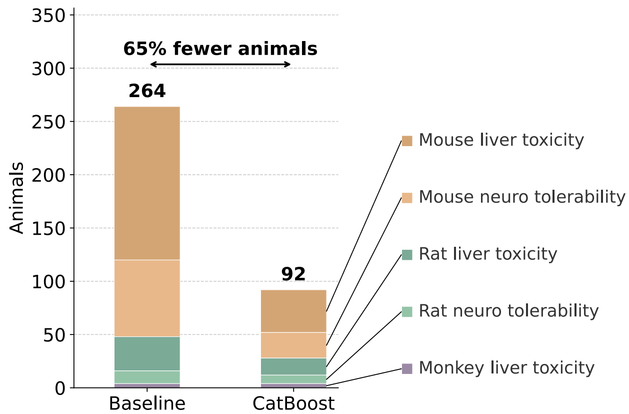

Figure A5: Animal usage reduction under CatBoost pre-screening. Stacked bars show the number of animals required at each in vivo pipeline stage (assuming 4 animals per study). CatBoost pre-screening reduces the number of compounds entering animal studies, cutting total animal usage from the baseline.

A.5 Extraction Schema Reference

Table A2, Table A3, and Table A4 list the column definitions supplied to the Stage 3 LLM extraction pass for each assay category. Each schema was provided as a dictionary mapping column names to natural-language descriptions; the LLM generated a per-table mapping function that translated heterogeneous patent table formats into these target columns, performing unit conversions as specified. The dose-response schema is identical to in vitro inhibition except that dosage is extracted in the original reported unit (nM or μM) rather than normalised.

Table A2: Stage 3 extraction schema for in vitro inhibition and dose response. Dose response uses the same fields but extracts dosage in the reported unit (nM or μM) rather than normalising to nM.

Field	Type	Description
Inhibition_pct	float	Percent reduction in target RNA vs untreated control (UTC).
dosage_nm	float	ASO concentration in nM.
target_RNA	string	mRNA target name, without suffixes (e.g. "HBV" not "HBV mRNA").
cell_line	string	Cell line identifier (e.g. HCT116, A431).
cell_line_species	string	Species of the cell line (human, mouse, etc.).
transfection_method	string	Electroporation, Gymnosis, or Lipofection.
treatment_period_hrs	float	Hours between transfection and inhibition measurement.
cells_per_well	float	Number of cells seeded per well.

Table A3: Stage 3 extraction schema for hepatorenal toxicity. Only rows with matching units are retained; non-convertible values are excluded by the extraction function.

Field	Type	Description
ALT	float	Alanine aminotransferase, ALT (IU/L).
AST	float	Aspartate aminotransferase, AST (IU/L).
ALB	float	Albumin, ALB (g/dL).
BUN	float	Blood urea nitrogen (mg/dL).
CREA	float	Creatinine (mg/dL).
TBIL	float	Total bilirubin (mg/dL). Aliases: bilirubin, BIL.
PC_ratio	float	Protein/creatinine ratio (unitless).
dosage	string	Dose as value + unit (e.g. "30 mg/kg").
species	string	Mouse, rat, or monkey.
species_strain	string	Animal strain (e.g. C57BL/6 mice).
num_doses	int	Number of doses administered before measurement.
dosing_period_days	float	Days between first administration and measurement.
measurement_source	string	Plasma, urine, or other.
administration_method	string	Route of administration (e.g. subcutaneous).
target_RNA	string	mRNA target name.

Table A4: Stage 3 extraction schema for **neurotoxicity**.

Field	Type	Description
FOB_score	list[float]	Functional observational battery score (0–7). bFOB for mouse, mFOB for rat.
dosage_ug	float	Dosage in μg. Converted from mg where reported (1 mg $= 1000 \mu g$).
species	string	Mouse or Rat.
species_strain	string	Animal strain (e.g. C57BL/6 mice, Sprague-Dawley rats).
latency_time_hours	float	Time between administration and measurement (hours).
administration_method	string	ICV (intracerebroventricular) or IT (intrathecal).
tolerability_score_type	string	Behavioural (bFOB) or Body parts (mFOB).

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state three contributions (cost-based evaluation framework, ASO Atlas 2.0 dataset, baseline benchmark) that directly correspond to the experimental results in Section 3 to Section 5. Limitations are discussed in Section 6.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: We discuss limitations in Section 6, including the reliance on chemistry-only features without target context, the single-organisation data source (Ionis Pharmaceuticals patents), and the binary classification approach that discards intermediate toxicity cases. No strong assumptions were made about the datasets or models benchmarked in this study.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [NA]

   Justification: The paper does not include formal theoretical results. The cost-based framework is an empirical model, not a theorem.

4. **Experimental Result Reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Section 3 to Section 5 fully describe the extraction pipeline, data cleaning, model features, cross-validation procedure, and cost framework with per-stage costs. All model configurations are specified.

5. **Open Access to Data and Code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: All code and data necessary to reproduce the main experimental results are openly available. The full codebase, including preprocessing, modeling, and evaluation scripts, is released at `https://github.com/ASO-Atlas/aso-atlas-2.0`. The underlying dataset is accessible at `https://huggingface.co/datasets/barneyhill/aso-atlas-2`.

6. **Experimental Setting/Details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

Justification: We provide an extensive description of the experimental setup in Section 3 to Section 5, including data splits, model features, label thresholds, and hyperparameter selection strategy. All code used to produce and visualise the results will be publicly released.

7. **Experiment Statistical Significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: All model metrics and cost-based evaluation results are reported as mean ± standard deviation across 5 GroupKFold cross-validation folds. The source of variability (fold-level variation) is stated.

8. **Experiments Compute Resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: The extraction pipeline LLM API cost ($500), OligoGym benchmark CPU runtime (2–4 hours), and OligoAI GPU training time (25 GPU-hours on NVIDIA RTX A6000) are reported in the paper.

9. **Code of Ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The research uses publicly available patent data and does not involve human subjects, deception, or dual-use risks.

10. **Broader Impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: The paper discusses the positive societal impact of reducing animal use and preclinical screening costs. No direct negative societal impacts are foreseen for a benchmark dataset.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [NA]

    Justification: The paper poses no such risks. The dataset is derived from publicly available USPTO patent filings and the models are small classifiers for ASO property prediction.

12. **Licenses for Existing Assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

Justification: All existing assets are properly cited and their terms of use respected. Patent data is sourced from the USPTO bulk-data API and is in the public domain.

13. **New Assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: ASO Atlas 2.0 is documented with extraction pipeline details (Section 3.1), preprocessing and quality control procedures (Section A.2), and a dataset summary (Section 3). The code and data are released in a public repository with instructions for reproduction.

14. **Crowdsourcing and Research with Human Subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [NA]

    Justification: The paper does not involve crowdsourcing or research with human subjects.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [NA]

    Justification: The paper does not involve crowdsourcing or research with human subjects.

16. **Declaration of LLM Usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [Yes]

    Justification: The ASO Atlas 2.0 extraction pipeline uses GPT-5-mini (Stages 1–2) and GPT-5 (Stage 3) as core components, described in detail in Section 3.1.